Andres M. Arias Lorza, PhD

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Summary

Experienced managing biomedical image analysis, machine learning, and data analysis projects.

15 years experience supervising multidisciplinary teams of researchers in the development of biomedical imaging and data analysis research projects. 5 years experience in pharmaceutical research supervising projects related to clinical trials from new experimental therapies.

My overall knowledge of methodologies in imaging and data analysis, together with my technical knowledge in multiple programming languages and tools, and experience supervising research projects makes me an ideal candidate to manage and supervise research projects from its conception, development, validation, and manufacturing.

Experience

Associate Researcher, Zuse Institute Berlin – Berlin, DE — May 2022 – Present

Tutored PhD and masters students, also supervised several research projects related to medical imaging such as:

- Knee image segmentation using U-Nets and transfer learning
- Knee image reconstruction method for enabling tissue deformation analysis during motion

Postdoctoral Researcher, Moffitt Cancer Center – Tampa, FL — Jan 2018 – Apr 2022

Supervised several research projects related to clinical trials data analysis such as:

- Quantitative imaging tools to evaluate cancer treatment responses in clinical trials
- Identification of imaging biomarkers to predict treatment response with >70% accuracy

Doctoral Researcher, Erasmus University – Rotterdam, NL — Oct 2011 – Oct 2017

- Developed novel segmentation and registration methods for carotid artery imaging with >90% segmentation overlap and sub-millimeter registration accuracy
- Applied image analysis tools in big population studies resulting in discoveries in the mechanics of the arteries and stroke prediction

Education

Erasmus University, PhD in Biomedical Image Processing — Oct 2011 – Oct 2017

- Thesis: *Image analysis of the carotid artery: a (semi-)automatic approach*

Eindhoven University of Technology, Master in Embedded Systems — Aug 2009 – Sep 2011

- Thesis: *Analysis of 3D MRI Blood-Flow Data using Helmholtz Decomposition*

Pontificia Universidad Javeriana, BS in Electronics Engineering — Jan 2001 – Oct 2007

- Thesis: *Voice recognition system implemented in a Digital Signal Processor (DSP)*

Selected Publications

Magnetic resonance imaging of tumor response to stroma-modifying pegvorhyaluronidase alpha (PEGPH20) therapy in early-phase clinical trials — 2024

Arias et al., Nature Scientific Reports, 10.1038/s41598-024-62470-9

Dose-response assessment by quantitative MRI in a phase 1 clinical study of the anti-cancer vascular disrupting agent crolibulin — 2020

Arias et al., Nature Scientific Reports, 10.1038/s41598-020-71246-w

Carotid Artery Wall Segmentation in Multispectral MRI by Coupled Optimal Surface Graph Cuts — 2015

Arias et al., IEEE Transactions on Medical Imaging, 10.1109/TMI.2015.2501751

Projects

HR segmentation of LR images from public challenge datasets GitHub

- Development of a software tool to obtain high resolution segmentation of structures from low resolution and poor contrast images
- Tools Used: Batch, Python, Jupyter notebooks, AWS

Diffusion and perfusion parameters from MRI images GitHub

- Development a software tool that generates ADC maps from DW-MRI data; T1 maps from T1-MRI; and Ktrans; Vp, Ve, iAUC maps from DCE-MRI
- Tools Used: Matlab, multi-core programming

Software tool to analyze the knee movement from dynamic imaging Technical Report

- Development of a software tool to obtain high resolution reconstruction of dynamic low resolution images of the knee making possible to analyze bones and soft tissue deformations during movement
- Tools Used: Python, Matlab, high-performance computing programming

Software tool to segment the carotid arteries from MRI Technical Report

- Development of a software tool to segment the arteries from MRI using a graph-cup approach
- Tools Used: C++, Python, Matlab, high-performance computing programming

Technologies

Languages: C++, Python, Matlab, Batch, high-performance computing programming, HTML

Technologies: Amira, Mevislab, 3D slicer, Wolfram Mathematica, Amazon Web Services, Jupyter Notebooks, Weka machine learning tool, Elastix image registration tool

Courses & Certifications

Machine Learning Operations (MLOps) Coursera Online Course

Intro to TensorFlow for Deep Learning Udacity Online Course

Deep Learning with MATLAB Mathworks Online Course

Medical Imaging Summer School Favignana, IT

Scientific English Writing Rotterdam, NL

Computer Vision Summer School Ragusa, IT

Biomedical Image Analysis Summer School Paris, FR

Advanced Pattern Recognition Delft, NL

Knowledge Driven Image Segmentation Leiden, NL

Front end Vision & Multi-Scale Image Analysi Eindhoven, NL

ISMRM conference X 2 Virtual Meeting

ISMRM conference Montreal, CA

MICCAI conference Nagoya, JP

MICCAI conference Nice, FR